Jorge Chavez

Joliet, IL | j.chavez3019@gmail.com | (815) 351-0478 | www.jorgechavezuiuc.com

EDUCATION

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

- Master of Science in Electrical and Computer Engineering - GPA: 3.83/4.0 — May 2025
  - *Concentration in Machine Learning Verification*
- Bachelor of Science in Computer Engineering - GPA: 3.83/4.0 — May 2023

PROJECTS

$\alpha\beta$-CROWN, Neural Network Verification – PyTorch Based Verifier — Jan 2024 – Present

- Engineered a branch-and-bound optimization framework that **cut subproblems by 96%** and **accelerated verification by up to 85%**, driving the team to **1st place in VNN-COMP 2024 & 2025 (#1 on all 21 benchmarks)**, with all experiments deployed on standardized AWS A10G GPU instances to ensure reproducibility and cloud-scalable evaluation.
- Implemented a PGD-based soundness check integrated into the project's CI/CD pipeline to **automate validation of new contributions**, with support for local runs to streamline development.
- Optimized parallel execution in the branch-and-bound algorithm for complex queries, **achieving a 2x speedup on several benchmarks**.

Bounding Neural Network Implicit Surfaces – PyTorch — Sept 2024 – Present

- Developed **Guaranteed Inner-and-Outer Meshes (GIOM)**, a framework combining neural network verification with neural implicit surfaces to accelerate geometric queries in computer graphics.
- Achieved **3× faster rendering with 5 dB PSNR improvement**, **5× faster collision detection**, and **10× lower constructive solid geometry (CSG) reconstruction errors** compared to prior methods with minimal computational overhead.
- Integrated the open-source, state-of-the-art neural verification tool ($\alpha\beta$-CROWN) to produce certified bounding meshes, enabling **sound and efficient ray intersection, collision detection, and constructive solid geometry operations**.
- **Co-authored research paper** *(submission to ICLR 2026)*, contributing theoretical guarantees and practical benchmarks for high-fidelity neural surface rendering.

WORK EXPERIENCE

Motorola Solutions | Embedded Systems Engineer — May 2022 – Aug 2022

Schaumburg, IL

- Engineered a multi-threaded Python application for a custom software-defined radio (SDR) to monitor, classify, and replay radio signals as decoys, enhancing system security.
- Designed a memory-efficient signal processing pipeline using Welch's method**, cutting storage requirements from 2 MB/min to 0.015 MB/min (~99% savings)**, enabling long-term online recording and prediction.
- Enhanced communication reliability by applying digital signal processing and predictive analytics for signal recurrence and pattern detection.

LEADERSHIP EXPERIENCE

Illini Space Society | Controls Programmer and Project Lead — Sept 2025 – May 2025

- Supervised a 3-member team to design and deploy feedback controllers for autonomous systems using PyTorch (NNs) and PyBullet (simulation), enabling robust and effective trajectory stabilization.
- Applied both ML-based and classical approaches to achieve stable autonomous behavior: engineered neural-network controllers with Lyapunov stability guarantees (verified via αβ-CROWN) for inverted pendulum/cart-pole systems, and implemented LQR to achieve stable quadrotor control at arbitrary equilibrium points.
- Presented technical workshops on state-space modeling, feedback control, and Kalman filtering, mentoring 15 peers and strengthening the org's expertise in control theory.

RESEARCH AND SCHOLARLY WORK

ASSURED AND TRUSTWORTH AI RESEARCH LAB (ASTRAL)

- **Clip-and-Verify: Linear Constraint-Driven Domain Clipping for Accelerating Neural Network Verification**
  Neural Information Processing Systems (NeurIPS), 2025 - accepted, equal contribution
- **Guaranteed Bounding Meshes Extraction from Neural Implicit Surfaces via Neural Network Verification**
  International Conference on Learning Representations (ICLR), 2026 - under review, equal contribution
- **A Linear Constraint Driven Approach to Efficiently Enhancing Branch and Bound in Neural Network Verification**
  Master's Thesis, UIUC, 2025

Programming Skills: C/C++, Python, Matlab, Verilog, CUDA, PyTorch, Angular, OpenCV, Pandas, ROS